

7. Project Documentation Phase

ComicCraft – AI Comic Story Creator Using Gemini Models

Recording the project concept, architecture, implementation, and usage

1. Documentation Purpose

Project documentation explains what ComicCraft does, how its components work together, how to run it locally, and how users interact with the comic-generation workflow.

2. Project Summary

ComicCraft is a web-based AI comic creator. Users provide a prompt and preferences; the application generates a five-panel outline, narration and dialogue, illustrations, a preview, and a downloadable PDF.

3. Technology Stack

The supplied guide identifies FastAPI and Uvicorn for the backend/server, HTML/CSS/Jinja2 for the interface, Gemini Flash and Gemini Pro for text generation, Hugging Face Diffusers with Stable Diffusion for image generation, and FPDF for PDF export.

4. Module Reference

gemini_flash.py: generate_outline(). gemini_pro.py: generate_story(). image_generator.py: generate_image(). layout_builder.py: build_comic_layout(). exporters.py: save_pdf(). routes.py: application routes and workflow integration.

5. Route Reference

/ loads the homepage; /generate handles browser form submissions; /generate-comic/json handles JSON API requests; /export-success displays export confirmation; /test-image tests image generation.

6. User Guide

Open the local homepage, enter a story prompt and preferences, and submit the form. Review the generated comic on the preview page. Select the PDF download action to save the comic, then use the success page to return and create another comic.

7. Setup and Run Instructions

Create and activate a virtual environment, install dependencies from requirements.txt, configure GEMINI_API_KEY and HF_API_KEY, then run `uvicorn app.main:app --reload`. Open 127.0.0.1:8000 for the app and /docs for interactive API documentation.

8. Output and Storage

Generated panel images are stored under static/panels. Exported PDF files are stored under static/exports. The guide describes timestamped PDF filenames.

9. Scope and Future Enhancements

The current version focuses on generating a single comic and does not include user account management. The guide identifies user profiles, comic libraries, and multi-page story arcs as possible future enhancements.

10. Documentation Outcome

A complete project document set should include the project overview, requirements, architecture, module descriptions, route details, setup instructions, user workflow, and testing observations. Any test results should reflect actual execution rather than assumptions.